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Elliptic regularity on a pizza slice

This appendix solves the Dirichlet eigenvalue problem for the Laplacian on a circular sector of arbitrary aperture by separation of variables, and identifies the corner singularity that limits the convergence rate of the finite element method. The aperture $\frac{3}{2}\pi$ used in Theorem 6.13 is worked out in Example 2, and the second section applies the same technique to an anisotropic problem on a cylinder. For an aperture $\omega \in (0, 2\pi)$ we consider the Dirichlet eigenvalue problem

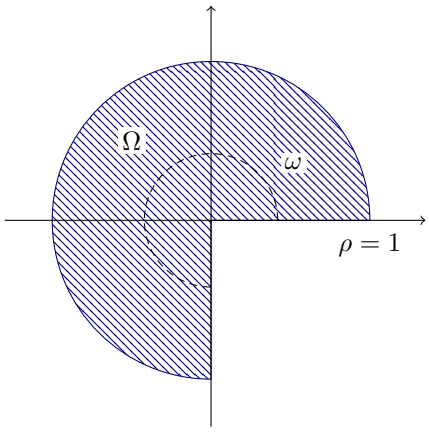
$$\begin{cases} \Delta\Phi(\rho,\theta) = -\lambda\,\Phi(\rho,\theta) & \text{in } \Omega, \\ \Phi(\rho,\theta) = 0 & \text{on } \partial\Omega, \end{cases} \tag{1}$$

on the circular sector of radius 1 and aperture ω,

$$\overline{\Omega} = \Big\{ \rho e^{i\theta} \in \mathbb{C} \,:\, \rho \in [0,1],\ \theta \in [0,\omega] \Big\}, \tag{2}$$

drawn in Figure 1. We impose the further condition $|\Phi(\rho,\theta)| < \infty$, which is dictated by the physics of the problem. The corner at the origin is re-entrant exactly when $\omega > \pi$.

Figure 1: The circular sector (2) of radius 1 and aperture ω on which the Dirichlet eigenvalue problem (1) is posed, drawn here for $\omega = \frac{3}{2}\pi$. For $\omega > \pi$ the corner at the origin is re-entrant, and it is what limits the elliptic regularity of the eigenfunctions.



Assume Φ depends separately on the radius and the angle, $\Phi(\rho,\theta) = \Theta(\theta)R(\rho)$. Writing the Laplacian in polar coordinates,

$$\Delta\Phi(\rho,\theta) = \frac{1}{\rho}R'(\rho)\Theta(\theta) + \frac{1}{\rho^2}R(\rho)\Theta''(\theta) + R''(\rho)\Theta(\theta), \tag{3}$$

so that (1) becomes

$$\frac{1}{\rho}R'(\rho)\Theta(\theta) + \frac{1}{\rho^2}R(\rho)\Theta''(\theta) + R''(\rho)\Theta(\theta) = -\lambda R(\rho)\Theta(\theta). \tag{4}$$

Dividing by $R(\rho)\Theta(\theta)$ and multiplying by ρ^2 separates the variables,

$$-\frac{\Theta''(\theta)}{\Theta(\theta)} = \rho\,\frac{R'(\rho)}{R(\rho)} + \rho^2\,\frac{R''(\rho)}{R(\rho)} + \lambda\rho^2. \tag{5}$$

Since the left-hand side depends only on θ and the right-hand side only on ρ, both equal the same constant ν. The Dirichlet condition on the two straight edges $\theta = 0$ and $\theta = \omega$ then gives the one-dimensional eigenvalue problem

$$\Theta''(\theta) = -\nu\,\Theta(\theta) \quad \text{in } (0,\omega), \qquad \Theta(0) = \Theta(\omega) = 0, \tag{6}$$

whose solutions are

$$\nu_n = \alpha_n^2, \qquad \Theta_n(\theta) = \sin(\alpha_n\theta), \qquad \alpha_n = \frac{n\pi}{\omega}, \qquad n \geq 1. \tag{7}$$

Substituting back into (5) gives an ordinary differential equation for R,

$$\rho^2 R''(\rho) + \rho R'(\rho) + R(\rho)\big(\lambda\rho^2 - \alpha_n^2\big) = 0, \tag{8}$$

and the change of variable $z = \sqrt{\lambda}\,\rho$ turns (8) into Bessel's equation of order α_n,

$$z^2 R''(z) + zR'(z) + \big(z^2 - \alpha_n^2\big)R(z) = 0. \tag{9}$$

Its general solution is $R(z) = AJ_{\alpha_n}(z) + BY_{\alpha_n}(z)$. Since $\alpha_n > 0$, the Bessel function of the second kind behaves like $-\pi^{-1}\Gamma(\alpha_n)(2/z)^{\alpha_n}$ as $z \to 0$ [5], so the boundedness hypothesis forces $B = 0$. The Dirichlet condition on the arc, $R(1) = J_{\alpha_n}(\sqrt{\lambda}) = 0$, then shows that $\sqrt{\lambda}$ must be a positive zero of J_{α_n}. Writing $z_{n,m}$ for the m-th positive zero of J_{α_n}, the eigenpairs of (1) are

$$\Phi_{n,m}(\rho,\theta) = J_{\alpha_n}(z_{n,m}\rho)\,\sin(\alpha_n\theta), \qquad \lambda_{n,m} = z_{n,m}^2, \qquad n,m \geq 1. \tag{10}$$

There are no others, because the Θ_n are complete in $\mathcal{L}^2((0,\omega))$ and, for each fixed n, the Fourier–Bessel functions $J_{\alpha_n}(z_{n,m}\rho)$ are complete in $\mathcal{L}^2((0,1);\rho\,\mathrm{d}\rho)$ [5]. The first positive zero of J_α increases with α [5], so the smallest eigenvalue is

$$\lambda_1 = \lambda_{1,1} = z_{1,1}^2, \tag{11}$$

the square of the first positive zero of $J_{\pi/\omega}$.

The regularity of the eigenfunctions is governed by their behaviour as $\rho \to 0$. The power series of the Bessel function [5],

$$J_\alpha(z) = \Big(\frac{z}{2}\Big)^{\alpha} \sum_{k\geq 0} \frac{(-1)^k}{k!\,\Gamma(\alpha+k+1)} \Big(\frac{z}{2}\Big)^{2k}, \tag{12}$$

gives

$$\Phi_{n,m}(\rho,\theta) = C_{n,m}\,\rho^{\alpha_n}\sin(\alpha_n\theta) + \rho^{\alpha_n+2}\,g_{n,m}(\rho^2)\sin(\alpha_n\theta), \qquad C_{n,m} = \frac{1}{\Gamma(\alpha_n+1)}\Big(\frac{z_{n,m}}{2}\Big)^{\alpha_n}, \tag{13}$$

with $g_{n,m}$ an entire function. The second term is two powers of ρ smoother than the first, so $\Phi_{n,m}$ has exactly the regularity of $\psi_{\alpha_n} = \rho^{\alpha_n}\sin(\alpha_n\theta)$. This function is the imaginary part of $\zeta \mapsto \zeta^{\alpha_n}$, which is holomorphic on the sector. Hence ψ_{α_n} is harmonic, with $|\nabla\psi_\alpha|^2 = \alpha^2\rho^{2\alpha-2}$ and $|D^2\psi_\alpha|^2 = 2\alpha^2(\alpha-1)^2\rho^{2\alpha-4}$, where $|D^2\psi|$ is the Frobenius norm of the Hessian. On $\Omega_r = \Omega \cap B_r(\mathbf{0})$ this gives

$$\int_{\Omega_r} |\nabla\psi_\alpha|^2\,\mathrm{d}x = \omega\,\alpha^2 \int_0^r \rho^{2\alpha-1}\,\mathrm{d}\rho, \qquad \int_{\Omega_r} |D^2\psi_\alpha|^2\,\mathrm{d}x = 2\,\omega\,\alpha^2(\alpha-1)^2 \int_0^r \rho^{2\alpha-3}\,\mathrm{d}\rho. \tag{14}$$

The first integral is finite for every $\alpha > 0$, so every eigenfunction lies in $W^{1,2}(\Omega)$. The second is finite if and only if $\alpha \geq 1$: the integral on the right converges for $\alpha > 1$, and for $\alpha = 1$ the function $\psi_1 = \rho\sin\theta$ is linear, so that its Hessian vanishes. Since $\alpha_n = n\pi/\omega$, all eigenfunctions lie in $W^{2,2}(\Omega)$ when $\omega \leq \pi$.

When $\omega > \pi$ the family $n = 1$, with $\alpha_1 = \pi/\omega < 1$, does not. Computing the Gagliardo–Slobodeckij seminorm of ψ_{α_1} in the same way shows more precisely that

$$\Phi_{1,m} \in W^{1+\frac{\pi}{\omega}-\varepsilon,2}(\Omega) \quad \text{for every } \varepsilon > 0, \qquad \Phi_{1,m} \notin W^{2,2}(\Omega) \quad \text{if } \omega > \pi. \tag{15}$$

Expanding the solution of the Poisson problem in the eigenfunctions makes this precise. For $f \in \mathcal{L}^2(\Omega)$ let $u_0 \in W_0^{1,2}(\Omega)$ be the variational solution of

$$-\Delta u_0 = f \quad \text{in } \Omega, \qquad u_0 = 0 \quad \text{on } \partial\Omega. \tag{16}$$

Theorem 1 *Let Ω be the sector (2), let $f \in \mathcal{L}^2(\Omega)$, and let u_0 be the solution of (16).*

(i) *If $\omega \le \pi$, then $u_0 \in W^{2,2}(\Omega)$ and $\|D^2 u_0\|_{\mathcal{L}^2(\Omega)} \le \|f\|_{\mathcal{L}^2(\Omega)}$.*

(ii) *If $\pi < \omega < 2\pi$, let $\alpha = \alpha_1 = \pi/\omega$ and $S_1 = \eta(\rho)\,\rho^\alpha \sin(\alpha\theta)$, with η a smooth cut-off equal to 1 near $\rho = 0$ and to 0 near $\rho = 1$. Then*

$$u_0 - C_1 S_1 \;\in\; W^{2,2}(\Omega), \qquad C_1 = \frac{1}{\pi}\int_\Omega f\,\big(\rho^{-\alpha} - \rho^\alpha\big)\sin(\alpha\theta)\,\mathrm{d}x, \tag{17}$$

and

$$|C_1| \le \Big(\frac{\omega}{2(\omega^2-\pi^2)}\Big)^{1/2}\|f\|, \qquad \|D^2(u_0 - C_1 S_1)\| \le \|f\| + |C_1|\,\|\Delta S_1\|, \tag{18}$$

with all norms in $\mathcal{L}^2(\Omega)$.

Hence $\big(W_0^{1,2}(\Omega), W^{0,2}(\Omega), W^{2,2}(\Omega)\big)$ is a shift triplet for u_0 in the first case and for $u_0 - C_1 S_1$ in the second. The singular function S_1 belongs to $W^{1+\frac{\pi}{\omega}-\varepsilon,2}(\Omega)$ for every $\varepsilon > 0$ but not to $W^{2,2}(\Omega)$.

Proof. We expand u_0 in the eigenfunctions (10). Inner products and norms without a subscript are those of $\mathcal{L}^2(\Omega)$. Let $\varphi_{n,m} = \Phi_{n,m}/\|\Phi_{n,m}\|$, which form a Hilbert basis of $\mathcal{L}^2(\Omega)$, and $\widehat{f}_{n,m} = (f, \varphi_{n,m})$. Then

$$u_0 = \sum_{n,m} \frac{\widehat{f}_{n,m}}{\lambda_{n,m}}\,\varphi_{n,m}.$$

The partial sums U_N of this series converge to u_0 in $W^{1,2}(\Omega)$, and $f_N = -\Delta U_N$ are the partial sums of the series of f.

We first prove an identity for one angular mode. Let $w(\rho,\theta) = v(\rho)\sin(\alpha\theta)$ with $\alpha = \alpha_n$, where v is smooth on $(0,1]$ and $v(1) = 0$. In polar coordinates

$$|D^2 w|^2 = \big(\partial_\rho^2 w\big)^2 + 2\Big(\partial_\rho \frac{\partial_\theta w}{\rho}\Big)^2 + \Big(\frac{\partial_\rho w}{\rho} + \frac{\partial_\theta^2 w}{\rho^2}\Big)^2,$$

and Δw is the sum of the first and of the last bracket. Since $\sin^2(\alpha\theta)$ and $\cos^2(\alpha\theta)$ have integral $\omega/2$ on $(0,\omega)$,

$$\int_0^\omega \big(|\Delta w|^2 - |D^2 w|^2\big)\rho\,\mathrm{d}\theta = \frac{\omega}{2}\Big(2v''\Big(v' - \frac{\alpha^2 v}{\rho}\Big) - 2\alpha^2\rho\Big(\frac{v'}{\rho} - \frac{v}{\rho^2}\Big)^2\Big) = \frac{\omega}{2}\,B'(\rho),$$

where $B(\rho) = v'^2 - 2\alpha^2\,v\,v'/\rho + \alpha^2 v^2/\rho^2$. Integrating over $r < \rho < 1$, with $B(1) = v'(1)^2 \ge 0$, gives

$$\int_{\Omega\setminus\Omega_r} |D^2 w|^2\,\mathrm{d}x \le \int_{\Omega\setminus\Omega_r} |\Delta w|^2\,\mathrm{d}x + \frac{\omega}{2}\,B(r). \tag{19}$$

The three brackets above are orthogonal on $(0,\omega)$ for two different angular modes. Hence (19) holds for a finite sum $w = \sum_n v_n(\rho)\sin(\alpha_n\theta)$, with B replaced by the sum of the corresponding B_n, and

$$\|D^2 w\| \le \|\Delta w\| \qquad \text{if } B_n(r) \to 0 \text{ as } r \to 0 \text{ for every } n. \tag{20}$$

By (13), a finite combination v of the functions $J_{\alpha_n}(z_{n,m}\rho)$, $m \geq 1$, satisfies $v = c\,\rho^\alpha + O(\rho^{\alpha+2})$ and $v' = c\,\alpha\,\rho^{\alpha-1} + O(\rho^{\alpha+1})$ as $\rho \to 0$, with $\alpha = \alpha_n$. Therefore

$$B(r) = 2c^2\alpha^2(1-\alpha)\,r^{2\alpha-2} + O(r^{2\alpha}), \tag{21}$$

which tends to 0 when $\alpha \geq 1$.

If $\omega \leq \pi$, then $\alpha_n \geq 1$ for every n, so (20) applies to U_N and to $U_N - U_M$:

$$\|D^2 U_N\| \leq \|f_N\| \leq \|f\|, \qquad \|D^2(U_N - U_M)\| \leq \|f_N - f_M\|.$$

Hence (U_N) is a Cauchy sequence in $W^{2,2}(\Omega)$, and its limit u_0 satisfies (i).

Let now $\pi < \omega < 2\pi$, so that $\alpha = \alpha_1 < 1 < \alpha_n$ for $n \geq 2$. The function

$$s(\rho,\theta) = \big(\rho^{-\alpha} - \rho^\alpha\big)\sin(\alpha\theta)$$

is harmonic in Ω and vanishes on $\partial\Omega$ away from the corner. It belongs to $\mathcal{L}^2(\Omega)$ because $\alpha < 1$, with

$$\|s\|^2 = \frac{\omega}{2}\int_0^1 \big(\rho^{-\alpha} - \rho^\alpha\big)^2 \rho\,\mathrm{d}\rho = \frac{\omega\,\alpha^2}{2(1-\alpha^2)},$$

and it is orthogonal to $\Phi_{n,m}$ for $n \geq 2$. For $n = 1$, Green's formula in $\Omega \setminus \Omega_r$ gives

$$-\lambda_{1,m}\int_{\Omega\setminus\Omega_r} \Phi_{1,m}\,s\,\mathrm{d}x = \int_{\Omega\setminus\Omega_r} \big(s\,\Delta\Phi_{1,m} - \Phi_{1,m}\,\Delta s\big)\,\mathrm{d}x = -\int_0^\omega \big(s\,\partial_\rho\Phi_{1,m} - \Phi_{1,m}\,\partial_\rho s\big)\Big|_{\rho=r} r\,\mathrm{d}\theta,$$

because s and $\Phi_{1,m}$ vanish on the rest of the boundary. By (13) the last integrand, including the factor r, equals $2\alpha\,C_{1,m}\sin^2(\alpha\theta) + O(r^2)$. Letting $r \to 0$ and using $\alpha\omega = \pi$,

$$(\Phi_{1,m}, s) = \frac{\pi\,C_{1,m}}{\lambda_{1,m}}. \tag{22}$$

Let $C_1 = \pi^{-1}(f,s)$, as in (17), and $C_1^N = \pi^{-1}(f_N,s)$. By (22), the component of U_N on the mode $n=1$ has a radial profile of the form $C_1^N\rho^\alpha + O(\rho^{\alpha+2})$. Moreover $|C_1 - C_1^N| \leq \pi^{-1}\|s\|\,\|f - f_N\|$ and $|C_1| \leq \pi^{-1}\|s\|\,\|f\|$, which is the first bound in (18) because $\alpha = \pi/\omega$.

Let $V_N = U_N - C_1^N S_1$. The radial profile v of its component on the mode $n=1$ vanishes at $\rho = 1$, and by the choice of C_1^N it satisfies $v = O(\rho^{\alpha+2})$ and $v' = O(\rho^{\alpha+1})$ near $\rho = 0$, where $\eta = 1$. Hence $B_1(r) \to 0$, while $B_n(r) \to 0$ for $n \geq 2$ by (21), and (20) applies to V_N and to $V_N - V_M$. The function ΔS_1 is bounded and vanishes near the corner, because $\rho^\alpha\sin(\alpha\theta)$ is harmonic. Therefore

$$\|D^2 V_N\| \leq \|f_N\| + |C_1^N|\,\|\Delta S_1\|, \qquad \|D^2(V_N - V_M)\| \leq \|f_N - f_M\| + |C_1^N - C_1^M|\,\|\Delta S_1\|.$$

So (V_N) is a Cauchy sequence in $W^{2,2}(\Omega)$. It converges to $u_0 - C_1 S_1$ in $W^{1,2}(\Omega)$, which proves (ii). The regularity of S_1 is that of ψ_{α_1}, given by (14) and (15). □

The singular function S_1 is ψ_{α_1} cut off away from the corner. For data in $W^{0,p}(\Omega)$ with $p \neq 2$, and for general domains with corners, the decomposition involves one singular function $\eta(\rho)\,\rho^{m\pi/\omega}\sin(m\pi\theta/\omega)$ for every integer $m \geq 1$ with $m\pi/\omega < 2 - 2/p$, provided $2 - 2/p$ is not itself of this form; see [3, Chapter 4], [2] and, for systems, [4]. By (15), when $\omega > \pi$ the first eigenfunction $\Phi_{1,1}$ is no smoother than $W^{1+\frac{\pi}{\omega}-\varepsilon,2}(\Omega)$. On quasi-uniform meshes of size h the best approximation error by piecewise linear elements in H^1 is then $O(h^{\pi/\omega-\varepsilon})$ rather than $O(h)$. Theorem 6.7 squares this error and predicts an eigenvalue convergence rate of $2\pi/\omega$, below the rate 2 obtained when $\omega \leq \pi$.

Example 2 For the sector of Theorem 6.13, $\omega = \frac{3}{2}\pi$ and (7) gives the orders $\alpha_n = \frac{2}{3}n$, so that

$$\Phi_{n,m}(\rho,\theta) = J_{\frac{2}{3}n}(z_{n,m}\rho)\,\sin\big(\tfrac{2}{3}n\theta\big), \qquad \lambda_{n,m} = z_{n,m}^2. \tag{23}$$

By (11) the smallest eigenvalue is $\lambda_1 = z_{1,1}^2$, where $z_{1,1} \approx 3.375610652$ is the first positive zero of $J_{2/3}$. This is the exact value used in Table 4.

Only the family $n = 1$ has $\alpha_n < 1$. For $n = 3k$ the order $\alpha_n = 2k$ is an integer, and $\Phi_{3k,m}$ is smooth up to the corner, because $J_{2k}(z\rho)e^{2ki\theta}$ is a smooth function on the whole plane. For the remaining $n \geq 2$ the order is a non-integer larger than 1, and $\Phi_{n,m}$ lies in $W^{2,2}(\Omega)$ without being smooth. For $n = 1$, (13) gives

$$\Phi_{1,m}(\rho,\theta) = C_{1,m}\,\rho^{\frac{2}{3}}\sin\!\big(\tfrac{2}{3}\theta\big) + O\big(\rho^{\frac{8}{3}}\big) \qquad \text{as } \rho \to 0, \tag{24}$$

and (15) says that $\Phi_{1,m}$ lies in $W^{\frac{5}{3}-\varepsilon,2}(\Omega)$ for every $\varepsilon > 0$ but not in $W^{2,2}(\Omega)$. In Theorem 1 the singular function is $S_1 = \eta(\rho)\,\rho^{\frac{2}{3}}\sin(\frac{2}{3}\theta)$. The best approximation error by piecewise linear elements in H^1 is therefore $O(h^{\frac{2}{3}-\varepsilon})$, and the predicted eigenvalue convergence rate is $2\pi/\omega = \frac{4}{3}$. The computed rates in Table 4 decrease towards this value.

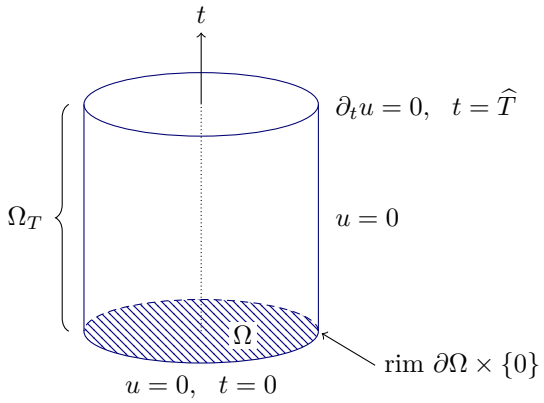
Elliptic regularity on a cylinder

Let us now consider the disk $\Omega = B_{R_0}(\mathbf{0}) \subset \mathbb{R}^2$ and the cylinder $\Omega_T = \Omega \times (0,\widehat{T})$, drawn in Figure 2, and discuss the elliptic regularity of the solution of the following problem, where $\varepsilon > 0$ and Δ_x is the Laplacian in the variable $x \in \Omega$:

$$\begin{cases} -\Delta_x u - \varepsilon\,\partial_t^2 u = f & \text{in } \Omega_T, \\ u(x,0) = 0 & \forall x \in \Omega, \\ u(x,t) = 0 & \forall x \in \partial\Omega,\ t \in (0,\widehat{T}), \\ \partial_t u(x,\widehat{T}) = 0 & \forall x \in \Omega. \end{cases} \tag{25}$$

The radius and the final time are written R_0 and $\widehat{T}$ because R and T denote, below, two factors of a separated solution, and the radial coordinate is ρ as in the previous section.

Figure 2: The cylinder $\Omega_T = \Omega \times (0,\widehat{T})$ over the disk $\Omega = B_{R_0}(\mathbf{0})$ on which problem (25) is posed. The solution vanishes on the bottom and on the lateral boundary, and its derivative in t vanishes on the top. The two conditions $u = 0$ meet at a right angle along the rim $\partial\Omega \times \{0\}$.



As before we study the problem via the eigenvalue problem associated to (25), posed in cylindrical coordinates (ρ,θ,t). Let $\Phi(\rho,\theta,t) = R(\rho)\Theta(\theta)T(t)$ and consider the eigenvalue problem

$$-\Delta_x \Phi - \varepsilon\,\partial_t^2 \Phi = \lambda\,\Phi \qquad \text{in } \Omega_T, \tag{26}$$

with the boundary conditions of (25). For a product it reads, before and after dividing by $R(\rho)\Theta(\theta)T(t)$,

$$-T\,\Delta_x(R\Theta) - \varepsilon\,T''R\Theta = \lambda\,R\Theta T, \qquad -\frac{\Delta_x(R\Theta)}{R\Theta} - \varepsilon\,\frac{T''}{T} = \lambda. \tag{27}$$

The first term depends only on (ρ,θ) and the second only on t, so by the standard separation of variables argument both terms of the sum must be constant. Therefore $R(\rho)\Theta(\theta)$ and $T(t)$ need to solve the eigenvalue problems

$$-\Delta_x\big(R(\rho)\Theta(\theta)\big) = \mu\, R(\rho)\Theta(\theta) \quad \text{in } \Omega, \qquad R(R_0) = 0, \tag{28}$$

$$-\varepsilon\, T''(t) = \zeta\, T(t) \quad \text{in } (0,\widehat{T}), \qquad T(0) = T'(\widehat{T}) = 0, \tag{29}$$

with $\lambda = \mu + \zeta$. We can solve either problem first. For simplicity we begin from (28), and observe that solving it was the main focus of the previous section.

Problem (28) is the problem (1) of the previous section with one change. On the full disk the half-lines $\theta = 0$ and $\theta = 2\pi$ coincide and carry no boundary condition. Instead Θ must be 2π-periodic, so (6) is replaced by

$$\Theta''(\theta) = -\nu\,\Theta(\theta), \qquad \Theta(\theta + 2\pi) = \Theta(\theta), \tag{30}$$

whose solutions are $\nu_n = n^2$ and $\Theta_n(\theta) = e^{in\theta}$ with $n \in \mathbb{Z}$. Equation (8) then becomes Bessel's equation of integer order $|n|$ in the variable $z = \sqrt{\mu}\,\rho$. Boundedness at the origin discards $Y_{|n|}$ as before, the singularity being logarithmic for $n = 0$, and the Dirichlet condition $R(R_0) = J_{|n|}(\sqrt{\mu}\,R_0) = 0$ shows that $\sqrt{\mu}\,R_0$ is a positive zero of $J_{|n|}$. Writing $j_{n,m}$ for the m-th positive zero of J_n and using $J_{-n} = (-1)^n J_n$, the eigenpairs of (28) are

$$W_{n,m}(\rho,\theta) = J_n(z_{n,m}\rho)\, e^{in\theta}, \qquad \mu_{n,m} = z_{n,m}^2, \qquad z_{n,m} = \frac{j_{|n|,m}}{R_0}, \qquad n \in \mathbb{Z},\ m \ge 1. \tag{31}$$

Their real and imaginary parts, $J_n(z_{n,m}\rho)\cos(n\theta)$ and $J_n(z_{n,m}\rho)\sin(n\theta)$, are the real eigenfunctions. There are no others, by the completeness of the trigonometric system and of the Fourier–Bessel functions [5]. Because the order is an integer, $W_{n,m}$ is a smooth function on the whole plane, as already observed in Example 2, and in particular $R(\rho)\Theta(\theta)$ is smooth on $\overline{\Omega}$.

We can then solve the second eigenvalue problem. The solutions of $-\varepsilon\,T'' = \zeta\,T$ with $T(0) = 0$ are the multiples of $\sin(\beta t)$, with $\zeta = \varepsilon\beta^2$ and $\beta > 0$; for $\zeta \le 0$ only $T = 0$ satisfies both boundary conditions. The condition $T'(\widehat{T}) = \beta\cos(\beta\widehat{T}) = 0$ requires $\beta\widehat{T}$ to be an odd multiple of $\pi/2$, hence

$$T_s(t) = \sin(\beta_s t), \qquad \beta_s = \frac{(2s-1)\pi}{2\widehat{T}}, \qquad \zeta_s = \varepsilon\,\beta_s^2 = \varepsilon\,\frac{(2s-1)^2\pi^2}{4\widehat{T}^2}, \qquad s \ge 1, \tag{32}$$

with $T_s'(t) = \beta_s\cos(\beta_s t)$ and $T_s''(t) = -\beta_s^2\,T_s(t)$.

The overall eigenfunctions are therefore given by the smooth sequence of functions

$$\Phi_{n,m,s}(\rho,\theta,t) = J_n(z_{n,m}\rho)\, e^{in\theta}\sin(\beta_s t), \qquad \lambda_{n,m,s} = z_{n,m}^2 + \varepsilon\,\beta_s^2, \tag{33}$$

for $n \in \mathbb{Z}$ and $m, s \ge 1$; the real eigenfunctions are $\operatorname{Re}\Phi_{n,m,s}$ and $\operatorname{Im}\Phi_{n,m,s}$. The functions (33) are bounded by 1 in modulus and orthogonal for the inner product $(f,g)_{\mathcal{L}^2(\Omega_T)} = \int_{\Omega_T} f\,\overline{g}\,\mathrm{d}x\,\mathrm{d}t$, but they are not normalised in $\mathcal{L}^2(\Omega_T)$, since [5]

$$\|\Phi_{n,m,s}\|^2_{\mathcal{L}^2(\Omega_T)} = \frac{\pi R_0^2\,\widehat{T}}{2}\, J_{|n|+1}\big(j_{|n|,m}\big)^2, \tag{34}$$

which tends to zero as $m \to \infty$, because the Bessel factor on the right behaves like $(2/(\pi j_{|n|,m}))^{1/2}$.

Lemma 3 *The solution operator $S\colon \mathcal{L}^2(\Omega_T) \to \mathcal{L}^2(\Omega_T)$, $Sf = u$, of problem* (25) *is compact and self-adjoint. Hence $\mathcal{L}^2(\Omega_T)$ has a Hilbert basis made of eigenfunctions of* (26) *[1, Chapter 6].*

Proof. Let V be the space of the functions in $W^{1,2}(\Omega_T)$ that vanish on $\Omega\times\{0\}$ and on $\partial\Omega\times(0,\widehat{T})$. The weak form of (25) is to find $u \in V$ such that

$$\int_{\Omega_T}\big(\nabla_x u\cdot\nabla_x\overline{v} + \varepsilon\,\partial_t u\,\partial_t\overline{v}\big)\,\mathrm{d}x\,\mathrm{d}t = \int_{\Omega_T} f\,\overline{v}\,\mathrm{d}x\,\mathrm{d}t \qquad \forall v \in V. \tag{35}$$

The form on the left is Hermitian and, by the Poincaré inequality, coercive on V, so S is well defined, self-adjoint and injective. It is compact because V is compactly embedded in $\mathcal{L}^2(\Omega_T)$. □

The functions (33) are such a basis once normalised, and there are no further eigenfunctions, because the $W_{n,m}$ are complete in $\mathcal{L}^2(\Omega)$ and the T_s are complete in $\mathcal{L}^2((0,\widehat{T}))$. Now this means that, numbering the eigenpairs (33) as $(\Phi_\ell,\lambda_\ell)_{\ell\ge1}$ with $\lambda_1\le\lambda_2\le\dots$, every function in $\mathcal{L}^2(\Omega_T)$ can be represented by a series in the Φ_ℓ. Using the representation for both f and u,

$$f=\sum_{\ell=1}^{\infty} f_\ell\,\Phi_\ell,\qquad u=\sum_{\ell=1}^{\infty} u_\ell\,\Phi_\ell,\qquad f_\ell=\frac{(f,\Phi_\ell)_{\mathcal{L}^2(\Omega_T)}}{\|\Phi_\ell\|^2_{\mathcal{L}^2(\Omega_T)}},\tag{36}$$

and taking $v=\Phi_\ell$ in (35), we obtain that (25) is verified if and only if the identity on the left of

$$u_\ell=\frac{f_\ell}{\lambda_\ell},\qquad \|\Phi_\ell\|_{W^{2,\infty}(\Omega_T)}\le C\,\lambda_\ell,\tag{37}$$

holds for every ℓ. The bound on the right holds with a constant C that depends only on ε, R_0 and $\widehat{T}$, and can be proven by direct computation. For a fixed $z>0$ let $W_\nu(\rho,\theta)=J_\nu(z\rho)\,e^{i\nu\theta}$, so that $\Phi_{n,m,s}=W_n\sin(\beta_s t)$ with $z=z_{n,m}$. Since $\partial_x\pm i\partial_y=e^{\pm i\theta}\big(\partial_\rho\pm\frac{i}{\rho}\partial_\theta\big)$, the recurrence relations $J'_\nu(z)\mp\frac{\nu}{z}J_\nu(z)=\mp J_{\nu\pm1}(z)$ [5] give

$$(\partial_x\pm i\partial_y)\big[J_\nu(z\rho)\,e^{i\nu\theta}\big]=\mp z\,J_{\nu\pm1}(z\rho)\,e^{i(\nu\pm1)\theta},\tag{38}$$

that is $(\partial_x\pm i\partial_y)W_\nu=\mp z\,W_{\nu\pm1}$. Then we have

$$\begin{gathered}\partial_x W_\nu=\tfrac{z}{2}\big(W_{\nu-1}-W_{\nu+1}\big),\qquad \partial_y W_\nu=\tfrac{iz}{2}\big(W_{\nu-1}+W_{\nu+1}\big),\qquad \partial_x\partial_y W_\nu=\tfrac{iz^2}{4}\big(W_{\nu-2}-W_{\nu+2}\big),\\ \partial_x^2 W_\nu=\tfrac{z^2}{4}\big(W_{\nu-2}-2W_\nu+W_{\nu+2}\big),\qquad \partial_y^2 W_\nu=-\tfrac{z^2}{4}\big(W_{\nu-2}+2W_\nu+W_{\nu+2}\big).\end{gathered}\tag{39}$$

For an integer order, $|J_n|\le1$ on the real axis, because $J_0^2+2\sum_{k\ge1}J_k^2=1$ [5]. This implies that the derivatives of Φ_ℓ of order at most two are bounded by 1, z_ℓ, β_ℓ, z_ℓ^2, $z_\ell\beta_\ell$ and β_ℓ^2, where z_ℓ and β_ℓ are the values of $z_{n,m}$ and β_s that belong to Φ_ℓ. Since $z_\ell^2=\lambda_\ell-\varepsilon\beta_\ell^2$, we have $z_\ell^2\le\lambda_\ell$, $\beta_\ell^2\le\lambda_\ell/\varepsilon$ and $z_\ell\beta_\ell\le\lambda_\ell/(2\sqrt{\varepsilon})$, and the lower order terms are bounded in the same way because $\lambda_\ell\ge\lambda_1>0$. This proves the bound in (37), with a constant C of order ε^{-1} as $\varepsilon\to0$.

The bound in (37) gives the following.

Proposition 4 *Let $f\in\mathcal{L}^2(\Omega_T)$ have coefficients (36) such that $\sum_{\ell=1}^{\infty}|f_\ell|<+\infty$. Then the solution u of (25) belongs to $C^2(\overline{\Omega_T})\subset W^{2,\infty}(\Omega_T)$, and*

$$\|u\|_{W^{2,\infty}(\Omega_T)}\le C\sum_{\ell=1}^{\infty}|f_\ell|<+\infty,\tag{40}$$

with the constant C of (37).

Proof. By (37),

$$\sum_{\ell=1}^{\infty}\|u_\ell\Phi_\ell\|_{W^{2,\infty}(\Omega_T)}\le\sum_{\ell=1}^{\infty}\frac{|f_\ell|}{\lambda_\ell}\,C\lambda_\ell=C\sum_{\ell=1}^{\infty}|f_\ell|,$$

so the series of u in (36) converges in $C^2(\overline{\Omega_T})$. □

Example 5 The summability assumption of Proposition 4 can fail for smooth data. When it holds, the series of f in (36) converges uniformly, so that f is continuous and vanishes on the bottom and on the lateral boundary, where every Φ_ℓ vanishes. A condition of this kind is necessary. Indeed, if $u\in C^2(\overline{\Omega_T})$ then $\Delta_x u=0$ on the bottom, where $u=0$, and $\partial_t^2u=0$ on the lateral boundary, where $u=0$. On the rim $\partial\Omega\times\{0\}$ both hold, so that $f=-\Delta_x u-\varepsilon\,\partial_t^2u=0$ there.

For $f\equiv1$ the failure is explicit. Only the radial modes enter, with $f_{0,m,s}=4/\big(\widehat{T}\,j_{0,m}\,\beta_s\,J_1(j_{0,m})\big)$, and these coefficients are not summable. Writing $u=\sum_m g_m(t)\,J_0(z_{0,m}\rho)$ and solving $-\varepsilon\,g_m''+z_{0,m}^2\,g_m=c_m$ with $g_m(0)=g_m'(\widehat{T})=0$ gives

$$u(\rho,t)=\sum_{m=1}^{\infty}\frac{c_m}{z_{0,m}^2}\left(1-\frac{\cosh\big(\kappa_m(\widehat{T}-t)\big)}{\cosh(\kappa_m\widehat{T})}\right)J_0(z_{0,m}\rho),\qquad \kappa_m=\frac{z_{0,m}}{\sqrt{\varepsilon}},$$

where $c_m = 2/\big(j_{0,m}J_1(j_{0,m})\big)$ are the Fourier–Bessel coefficients of 1. For $0 < t < \widehat{T}$ the series can be differentiated term by term, and on the lateral boundary

$$\partial_\rho \partial_t u(R_0, t) = -\frac{2}{\sqrt{\varepsilon}} \sum_{m=1}^{\infty} \frac{1}{j_{0,m}} \frac{\sinh\big(\kappa_m(\widehat{T}-t)\big)}{\cosh(\kappa_m \widehat{T})}. \tag{41}$$

Every term increases to $\tanh(\kappa_m\widehat{T})/j_{0,m}$ as $t \to 0$, and $\sum_m 1/j_{0,m}$ diverges because $j_{0,m}$ grows like πm. Therefore $\partial_\rho\partial_t u(R_0,t) \to -\infty$, at the rate $\frac{2}{\pi\sqrt{\varepsilon}}\log t$, and $u \notin W^{2,\infty}(\Omega_T)$, although f is smooth. This is the logarithmic singularity of a corner of aperture $\pi/2$, to which the rim reduces in the variables $(R_0 - \rho,\, t/\sqrt{\varepsilon})$.

For a general $f \in \mathcal{L}^2(\Omega_T)$ the expansion gives the following.

Proposition 6 *Let $f \in \mathcal{L}^2(\Omega_T)$. Then the solution u of (25) belongs to $W^{2,2}(\Omega_T)$, and*

$$\|D_x^2 u\| \le \|f\|, \qquad \|\nabla_x \partial_t u\| \le \frac{1}{2\sqrt{\varepsilon}}\,\|f\|, \qquad \|\partial_t^2 u\| \le \frac{1}{\varepsilon}\,\|f\|, \tag{42}$$

where $\|\cdot\|$ is the norm of $\mathcal{L}^2(\Omega_T)$ and $D_x^2 u$ is the Hessian of u in the variable x.

Proof. Let $\varphi_\ell = \Phi_\ell/\|\Phi_\ell\|$ and $\widehat{f_\ell} = (f,\varphi_\ell)_{\mathcal{L}^2(\Omega_T)}$, so that $u = \sum_\ell \lambda_\ell^{-1}\widehat{f_\ell}\,\varphi_\ell$ and $\sum_\ell |\widehat{f_\ell}|^2 = \|f\|^2$ by Lemma 3. Since $-\Delta_x\varphi_\ell = z_\ell^2\varphi_\ell$ and $-\partial_t^2\varphi_\ell = \beta_\ell^2\varphi_\ell$,

$$\|\Delta_x u\|^2 = \sum_\ell \Big(\frac{z_\ell^2}{\lambda_\ell}\Big)^2 |\widehat{f_\ell}|^2 \le \|f\|^2, \qquad \|\partial_t^2 u\|^2 = \sum_\ell \Big(\frac{\beta_\ell^2}{\lambda_\ell}\Big)^2 |\widehat{f_\ell}|^2 \le \frac{1}{\varepsilon^2}\,\|f\|^2.$$

Integrating by parts in x and in t, with the boundary conditions of (28) and (29), shows that the functions $\nabla_x\partial_t\varphi_\ell$ are orthogonal in $\mathcal{L}^2(\Omega_T)$, with $\|\nabla_x\partial_t\varphi_\ell\|^2 = z_\ell^2\beta_\ell^2$. Hence

$$\|\nabla_x \partial_t u\|^2 = \sum_\ell \frac{z_\ell^2\beta_\ell^2}{\lambda_\ell^2}\,|\widehat{f_\ell}|^2 \le \frac{1}{4\varepsilon}\,\|f\|^2,$$

because $z_\ell^2 \cdot \varepsilon\beta_\ell^2 \le \frac{1}{4}\big(z_\ell^2 + \varepsilon\beta_\ell^2\big)^2$. Finally, for $v \in W^{2,2}(\Omega) \cap W_0^{1,2}(\Omega)$ on the disk, integrating by parts twice gives [3, Chapter 3]

$$\int_\Omega |\Delta_x v|^2\,\mathrm{d}x = \int_\Omega |D_x^2 v|^2\,\mathrm{d}x + \frac{1}{R_0}\int_{\partial\Omega} |\partial_\nu v|^2\,\mathrm{d}\sigma,$$

so that $\|D_x^2 u\| \le \|\Delta_x u\| \le \|f\|$. All these identities are applied to the partial sums of u, which are smooth, and pass to the limit. □

In the terminology of Theorem 1, $\big(V, W^{0,2}(\Omega_T), W^{2,2}(\Omega_T)\big)$ is therefore a shift triplet for u, with V the space of Lemma 3. Only the constants of the derivatives in t degenerate as $\varepsilon \to 0$.

Example 7 In Proposition 6 the space $W^{2,2}(\Omega_T)$ cannot be replaced by $W^{2,\infty}(\Omega_T)$, even for bounded data that vanish on the bottom and on the lateral boundary. Let $v(\rho,\theta) = \rho^2\log\rho\,\sin 2\theta$, that is $v = 2xy\log\rho$ in the Cartesian coordinates (x,y) of (38). Then

$$\Delta_x v = 4\sin 2\theta, \qquad \partial_x\partial_y v = 2\log\rho + 2 - \sin^2 2\theta,$$

so $\Delta_x v$ is bounded while $\partial_x\partial_y v$ is not. Let χ be a smooth function of ρ with $\chi = 1$ for $\rho \le R_0/3$ and $\chi = 0$ for $\rho \ge 2R_0/3$, and let $u = \chi\, v\, T_1$, with $T_1(t) = \sin(\beta_1 t)$ as in (32). Then $u \in W^{2,p}(\Omega_T)$ for every $p < \infty$ and u satisfies the boundary conditions of (25), so it is the solution of (25) with datum $f = \big(-\Delta_x(\chi v) + \varepsilon\,\beta_1^2\,\chi v\big)\,T_1$. This f is bounded and vanishes on the bottom and near the lateral boundary. Nevertheless $u \notin W^{2,\infty}(\Omega_T)$, because $\partial_x\partial_y u = (2\log\rho + 2 - \sin^2 2\theta)\,T_1(t)$ for $\rho < R_0/3$. The datum is discontinuous on the axis $\rho = 0$. Some regularity of f in the space variable is therefore needed.

Two derivatives of f in $\mathcal{L}^2$ are enough, provided that f vanishes on the lateral boundary. The proof relies on the orthogonality of the eigenfunctions, as in Proposition 6, and not on the summability of the coefficients.

Proposition 8 *Let $f \in W^{2,2}(\Omega_T)$ vanish on the lateral boundary $\partial\Omega \times (0, \widehat{T})$. Then the solution u of (25) belongs to $W^{4,2}(\Omega_T) \subset C^{2,1/2}(\overline{\Omega_T}) \subset W^{2,\infty}(\Omega_T)$. Moreover*

$$\|D_x^2 \Delta_x u\| \le \|\Delta_x f\|, \qquad \|\nabla_x \partial_t \Delta_x u\| \le \frac{1}{2\sqrt{\varepsilon}} \|\Delta_x f\|, \qquad \|\partial_t^2 \Delta_x u\| \le \frac{1}{\varepsilon} \|\Delta_x f\|, \tag{43}$$

and $\|u\|_{W^{4,2}(\Omega_T)} \le C \|f\|_{W^{2,2}(\Omega_T)}$, with a constant C that depends only on ε, R_0 and $\widehat{T}$.

No condition is imposed on f on the bottom or on the top. For example $f = R_0^2 - |x|^2$ is admissible. It does not vanish on the bottom, so its coefficients are not summable, as observed in Example 5, and Proposition 4 does not apply.

Proof. For almost every t the function $f(\,\cdot\,, t)$ belongs to $W^{2,2}(\Omega) \cap W_0^{1,2}(\Omega)$. Since $W_{n,m}$ vanishes on $\partial\Omega$ and $-\Delta_x W_{n,m} = z_{n,m}^2 W_{n,m}$, Green's formula on the disk gives

$$(-\Delta_x f, \Phi_\ell)_{\mathcal{L}^2(\Omega_T)} = z_\ell^2 \, (f, \Phi_\ell)_{\mathcal{L}^2(\Omega_T)} \qquad \text{for every } \ell.$$

Only the variable x is integrated by parts, so nothing is required of f at $t = 0$ or at $t = \widehat{T}$. Hence the coefficients of $g = -\Delta_x f \in \mathcal{L}^2(\Omega_T)$, defined as in (36), are $g_\ell = z_\ell^2 f_\ell$. Let w be the solution of (25) with datum g. By (37) its coefficients are $w_\ell = z_\ell^2 f_\ell / \lambda_\ell = z_\ell^2 u_\ell$. By Proposition 6 the function $u(\,\cdot\,, t)$ also belongs to $W^{2,2}(\Omega) \cap W_0^{1,2}(\Omega)$ for almost every t, so the same formula shows that $z_\ell^2 u_\ell$ are the coefficients of $-\Delta_x u$. Therefore $-\Delta_x u = w$, and Proposition 6 applied to w gives $\Delta_x u \in W^{2,2}(\Omega_T)$ and (43). The equation then shows that

$$\varepsilon \, \partial_t^2 u = -\Delta_x u - f \in W^{2,2}(\Omega_T).$$

This controls the derivatives of u of order at most four with at least two derivatives in t. For the others, note that for almost every t the functions $u(\,\cdot\,, t)$ and $\partial_t u(\,\cdot\,, t)$ belong to $W_0^{1,2}(\Omega)$ and solve the Dirichlet problems

$$-\Delta_x u(\,\cdot\,, t) = w(\,\cdot\,, t) \in W^{2,2}(\Omega), \qquad -\Delta_x \partial_t u(\,\cdot\,, t) = \partial_t w(\,\cdot\,, t) \in W^{1,2}(\Omega).$$

The disk has a smooth boundary, so $u(\,\cdot\,, t) \in W^{4,2}(\Omega)$ and $\partial_t u(\,\cdot\,, t) \in W^{3,2}(\Omega)$, with norms bounded by those of the right-hand sides [3, Chapter 2]. Integrating in t gives $u \in W^{4,2}(\Omega_T)$. The bound on its norm follows because $\|w\|_{W^{2,2}(\Omega_T)} \le C \|\Delta_x f\|$, by Proposition 6 and the coercivity of (35). The inclusion in $C^{2,1/2}(\overline{\Omega_T})$ is the Sobolev embedding in dimension three [3, Chapter 1]. □

Remark 9 The two inclusions in Proposition 8 hold for every function in $W^{4,2}(\Omega_T)$ and use no property of f. The second derivatives of u belong to $W^{2,2}(\Omega_T)$, which is contained in $W^{1,6}(\Omega_T)$ by the Sobolev embedding, and $W^{1,6}(\Omega_T) \subset C^{0,1/2}(\overline{\Omega_T})$ by Morrey's embedding, because $1 - \frac{3}{6} = \frac{1}{2}$. The regularity $u \in W^{4,2}(\Omega_T)$ is a different matter. For the Dirichlet problem in a domain with a smooth boundary, the classical regularity theory gives $u \in W^{4,2}$ for every datum in $W^{2,2}$, with no condition on its boundary values [3, Chapter 2]. This fails in the cylinder, because the bottom and the lateral boundary meet at a right angle along the rim. The datum $f \equiv 1$ belongs to $W^{2,2}(\Omega_T)$, but by Example 5 its solution is not in $W^{2,\infty}(\Omega_T)$, hence not in $W^{4,2}(\Omega_T)$. More generally, if $u \in W^{4,2}(\Omega_T)$ then $u \in C^2(\overline{\Omega_T})$, and Example 5 shows that $f = 0$ on the rim. The classical theory still applies in every subdomain at a positive distance from the rim and from the edge $\partial\Omega \times \{\widehat{T}\}$ of the top. A condition on f is thus needed only because of the edges, and the one assumed in Proposition 8 is sufficient.

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